

Listen, Don’t Read: Repairing Chain-of-Thought Data for the DCASE 2026 Audio-Dependent QA Task

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Abstract—We describe our submission to DCASE 2026 Task 5, Audio-Dependent Question Answering (ADQA), an entry to the sub-10B lightweight champion track. The official training set pairs each multiple-choice question with a chain-of-thought (CoT) trace generated by Gemini 3.1 Pro, but our audit finds these traces systematically flawed: 68.5% never engage the answer options they should discriminate, and 10.1% are *contaminated*, reasoning from a written “transcript” that does not exist at inference time and teaching exactly the textual-hallucination behaviour the benchmark penalizes. We therefore repair the data rather than the model, regenerating the 14,129 flagged traces with a blind, audio-only Gemma-4-E4B-it→Qwen3-Omni-30B-A3B teacher cascade under rejection sampling: a trace is kept only when the teacher independently selects the gold option from the audio alone. We fine-tune Qwen2.5-Omni-7B on the repaired data and further optimize the best variant with GRPO; a reference-guided reasoning judge adds no accuracy, so our strongest model uses answer-accuracy and format rewards alone. Using only the 19,480 officially provided items and no external data or models, our best system scores 58.4% on the development set (+5.4 points over the base model) and 50.6% on the withheld ADQA-Bench evaluation set.

Index Terms—audio question answering, chain-of-thought distillation, training-data repair, reinforcement learning

1. INTRODUCTION

Large audio-language models (LALMs) increasingly answer questions about speech, sound, and music, yet much of their measured accuracy can come from language priors rather than from genuine listening. DCASE 2026 Task 5, Audio-Dependent Question Answering (ADQA), targets exactly this gap: every evaluation question passes an audio-dependency filtering (ADF) pipeline that removes items whose correct option can be recovered from the question and choices alone [1], and systems are ranked by top-1 accuracy on the 3,000-question ADQA-Bench evaluation set.

The official training corpus, AudioMCQ-StrongAC-GeminiCoT, provides 19,480 multiple-choice items drawn from the 570k+ sample AudioMCQ dataset [2]. Each item carries a chain-of-thought (CoT) trace, a written step-by-step rationale that ends in the answer, generated by Gemini 3.1 Pro. A deterministic audit of these traces (Section 2) reveals two pervasive defects. In 68.5% of items the trace never mentions at least one distractor (a wrong option), giving little supervision for the option-discrimination skill the benchmark stresses (Q1). In 10.1% it reasons from a written “transcript” that does not exist at inference time (Q2).

We repair the data and then measure what the repair buys. A *teacher* model regenerates each of the 14,129 flagged traces under *blind rejection sampling*: it is shown only the audio, the question, and the options, and its trace is kept only when the option it chooses matches the gold label, which is evidence that the teacher actually listened. Every kept trace must also give each option a perceptual verdict before it commits to an answer (examples in Section 2.4). We then train the *student*, Qwen2.5-Omni-7B, in two stages. Supervised fine-tuning (SFT) teaches it to imitate the traces, on three controlled data variants that differ only in which traces are repaired. Group Relative Policy Optimization (GRPO) [3], a reinforcement-learning

method, then samples several answers per question, rewards those that choose the correct option in the required format, and makes answers that score above their group’s average more likely. We run GRPO from both an unrepaired and a repaired SFT model, the latter also with a *reasoning judge*: a text-only language model that scores each sampled rationale against a reference rationale and adds that score to the reward.

We reuse the training scripts of the AudioMCQ recipe [2] but use only the officially provided items and no external training data, which, together with a sub-10B model, places our entry in the lightweight champion track. Our best system scores 50.6% on the official ADQA-Bench evaluation set. The controlled comparison separates two effects: the repair changes what the model says, not how often it is right (after SFT, answers that cite a non-existent transcript fall from 6.9% to 0.6% while accuracy is unchanged), whereas GRPO adds about 1.6 points from either initialization.

This paper contributes: (1) a reproducible audit of the official CoT labels; (2) a blind rejection-sampling regeneration pipeline; (3) a controlled SFT-and-GRPO study, completed after the challenge, that separates the effect of the repair from that of GRPO; and (4) a post-challenge analysis of the GRPO learning signal and the reasoning judge.

2. TRAINING-DATA AUDIT AND REPAIR

We audit all 19,480 Gemini CoT traces with a deterministic classifier, so every statistic below is reproducible.

2.1. Two defects in the official traces

Q1: the trace ignores the answer options. When a person answers a multiple-choice question about audio, they listen first and then judge every option: accept the one the audio supports, reject the ones it contradicts, and leave a verdict open when unsure, before making the final choice. A reasoning trace meant to teach this behaviour must therefore talk about the options. A trace is flagged Q1 when it fails to do so, that is, when at least one of the three distractor options is never mentioned anywhere in the trace (we search for each distractor’s exact wording, ignoring capitalization). Such a trace announces a winner without weighing the alternatives, and because ADQA’s distractors are text-plausible by construction (they survived perplexity-based soft filtering), telling them apart is the crux of the task. Engagement is in practice all-or-nothing: a trace that mentions any distractor almost always mentions all three; partial engagement (one or two of the three) is only 5.6% of items, and 12,249 of the 13,345 flagged traces mention no distractor at all. Q1 appears in *every* category (7,344 speech, 4,162 sound, 1,138 music, and 701 temporal traces), 13,345 rows in total (68.5%).

Q2: reference to a written document. A trace is flagged Q2 when it makes a determiner-led reference to a “transcript”, “transcription”, or “provided text” (exact regular expression). These traces reason from an artifact that does not exist at inference time, training the model to emit confident transcripts as if heard and to cite a document

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Table 1: Verified audit statistics per question type. The Q1 and Q2 columns overlap; the union counts each flagged row once.

type	<i>n</i>	clean	Q1	Q2	flagged (∪)
speech	10,505	2,408	7,344	1,919	8,097
sound	6,085	1,898	4,162	39	4,187
music	1,724	583	1,138	12	1,141
temporal	1,166	462	701	7	704
total	19,480	5,351	13,345	1,977	14,129

it was never given, precisely the textual-hallucination behaviour ADF penalizes. Unlike Q1, Q2 is almost exclusively a speech phenomenon: 1,919 of the 1,977 flagged rows (10.1% overall) are speech.

Conservative criteria. We adopt a candidate pattern only if every inspected match truly requires a written document; if even one printed context admitted a listening-compatible reading, the pattern was rejected. This deliberately leaves several suspicious surface forms unflagged: timestamps (“the receiver sound at 00:06” could be a genuinely heard event time or a copy from a transcript); quoted speech without a document reference; ambiguous words such as “the text”, “passage”, or “the prompt”; and traces that admit having no audio, which go to a 217-row manual-review sidecar because one audited case plausibly referred to a missing sound *within* a heard clip.

2.2. Illustrative traces

The excerpts below are copied verbatim from the training file; the labeler’s text is set in italics, our commentary in upright type. Q2 contamination concentrates in the speech subset:

- **Explicit transcript** (speech, SpeechCraft):
“The transcript clearly reads, ‘00:00 And there is only one of me, said the snapping turtle to himself.’”
 The trace names the document and copies it word for word, including a caption marker ‘00:00’ the speaker never uttered.
- **Lack-of-audio sidecar** (speech, SpeechCraft):
“Given my lack of audio input, I’m left to fall back on common patterns and typical datasets ... I choose ‘Male’...”
 The labeler admits it has no audio and answers from priors.

2.3. Blind rejection-sampling regeneration

We regenerate only the 14,129 flagged traces; the 5,351 clean rows keep their original CoT.

Blind. The teacher receives only the audio, the question, and the options. It never sees the gold answer or the original trace, so it has nothing to copy from.

Rejection sampling. The teacher must choose an option itself, and its trace is kept *only when that choice matches the gold label*: a correct blind answer is itself evidence of listening. We deliberately did not show the teacher the answer, because in a pilot the answer-conditioned teacher fabricated evidence, describing the same clip as having “trumpets” when shown the answer but “violins” when blind.

Accept filter. A kept trace must also pass a format check: it addresses every option by name with a perceptual verdict (for 231 traces we accepted paraphrased option names); it contains no document vocabulary; verbatim quotes span at most eight words; it is 30 to 320 words long; and it ends with the exact option text wrapped in an `<answer>` tag, the format later required by the answer-accuracy reward. A trace is rejected if the teacher’s choice is wrong or if any of these checks fails.

Design intent: deliberate per-option discrimination. Every regenerated trace follows an explicit elimination format: it opens with first-person perceptual evidence (“I hear ...”), then issues a separate verdict on *each* option by its exact wording, marking a distractor

Rejected with the perceptual reason it fails or Uncertain when the audio is inconclusive, before the final answer tag.

To control cost we run a teacher *cascade*. Gemma-4-E4B-it [4] (greedy) attempts every row and is accepted on 8,948 rows (63.3%); its 5,180 rejects escalate to the larger Qwen3-Omni-30B-A3B-Thinking [5], which together with sampling-retry fallbacks on the remainder recovers a further 3,935. Gemma is inexpensive but weak on fine perceptual distinctions such as speaker gender, instrument timbre, and sound identity, which is exactly where the larger model rescues it. The cascade accepts **12,883 of 14,129 regenerations (91.2%)**.

The remaining 1,246 rows, which no teacher answered correctly while blind even after the fallback retries, have no certifiable repair, since answer-conditioned regeneration of them was found to fabricate evidence, so we drop them entirely. This leaves a repaired pool of **18,234 items** that carry a verified CoT (5,351 clean plus 12,883 regenerated), every one drawn solely from the officially provided data. Dropping the same rows everywhere also removes any data-quantity confound, so the variants below differ only in CoT *content*.

2.4. From flagged to grounded

Two repairs (training-row id in parentheses) show the transformation; each regenerated trace reaches the same correct option but replaces guessing or transcript copying with first-person perception and a Rejected/Uncertain verdict per option. Excerpts are abridged and set in italics; the line breaks are ours.

Q1, speaker identity (librittsr_6482_289558)

Question: What best describes the speaker’s age and gender? **Correct option:** An older man.

Before (guesses from priors, names no other option): *“[...] am leaning towards ‘An older man’. It’s a common profile for this type of narration and aligns with typical datasets.”*

After: *“I hear a mid-to-low pitch suggesting an adult male.*

A teenage boy: Rejected (too mature);

A young woman: Rejected (clearly male);

An older man: Uncertain (cannot tell older from middle-aged);

An elderly woman: Rejected (male).”

Q2, spoken content (librittsr_7517_100437)

Question: Which author does the speaker specifically mention discovering? **Correct option:** Samuel Butler.

Before (answers from a written text that does not exist): *“Based on the provided text [...] ‘Samuel Butler’ immediately stands out.”*

After: *“I hear the speaker state he found Samuel Butler.*

Charles Dickens / Ernest Hemingway / George Eliot: Rejected (not mentioned);

Samuel Butler: stated explicitly.”

3. SYSTEMS

3.1. Model and data partition

All systems use Qwen2.5-Omni-7B [6] in a thinker-only configuration: the name counts only the 7.62B language backbone, and the deployed thinker adds a 0.64B audio encoder and a 0.68B vision encoder, for 8.93B inference parameters in total. The vision encoder is never exercised by this audio-only task, but we deploy the thinker unpruned and therefore count it. The speech-generation talker and its vocoder (a further 1.80B) are disabled because the task requires only text; this keeps the system within both the 30B per-component cap and the 10B threshold of the Lite-ADQA lightweight track. The 18,234-item repaired pool of Section 2 is split 70/30, by audio identity, into disjoint SFT and GRPO partitions, stratified by question type and

source dataset; because each audio clip contributes a single row, no clip is shared across the two. The assembled SFT training set has 12,569 items and the GRPO set 5,665 prompts, all from the provided corpus.

3.2. Supervised fine-tuning: three controlled variants

We run full-parameter SFT (no LoRA) on three variants that share the identical 12,569-row partition and differ only in which CoTs are repaired:

- **A, baseline:** every row keeps its original Gemini CoT (**0 repaired**). Retained as an unrepaired baseline and not submitted.
- **B, Q1+Q2:** every flagged row is replaced by a regenerated trace (**8,788 repaired**).
- **C, baseline+Q2:** only the Q2-bearing transcript rows are replaced by regenerated traces (**1,282 repaired**).

3.3. Reinforcement learning: GRPO

We further optimize variant C, the strongest SFT model on the development set (Section 5), with Group Relative Policy Optimization [3], initializing the model from the SFT-C checkpoint and training on the held-out 5,665-prompt partition of Section 3.1. For each prompt the model samples a group of 8 answers (rollouts) and each one receives a reward. GRPO then makes answers whose reward is above their group’s mean more likely and those below it less likely, while a small KL penalty keeps the model close to its SFT starting point. We train two runs that differ only in the reward. After the challenge we repeated the no-judge run from SFT-A, and from both SFT models with a second seed (Section 5).

Primary reward: accuracy and format. Our main system uses the simple reward

$$R = a + 0.5f, \quad (1)$$

where $a \in \{0, 1\}$ marks a correct final answer and $f \in \{0, 1\}$ marks a well-formed completion (a reasoning body and exactly one trailing answer tag).

Studied addition: a reference-guided reasoning judge. The second run adds a reasoning-quality term,

$$R_{\text{judge}} = a + 0.5f + 0.2a \frac{s-1}{4}, \quad (2)$$

where $s \in \{1, \dots, 5\}$ is a judge score, active only when $a = 1$. The judge is Prometheus-7b-v2.0 [7], a text-only language model trained to grade text on a 1-to-5 scale: it reads each rollout’s reasoning against the row’s reference CoT (the regenerated trace for flagged rows, the original Gemini CoT for clean rows) and never hears the audio, so it grades reasoning as text entailment rather than perception. Its intended role is to add variance where (1) has none: an all-correct group yields no gradient, and a fabricated-but-correct rollout is reinforced identically to a grounded one, so the judge should break these ties toward specific, reference-consistent, per-option reasoning. We cap its weight at 0.2, against 1.0 for accuracy and 0.5 for format, so the model optimizes the task rather than the judge.

4. EXPERIMENTAL SETUP

Hardware and framework. All training and inference ran on academic HPC clusters using NVIDIA H200 and H100 GPUs, with full-parameter tuning in ms-swift [8] under DeepSpeed ZeRO-2 and bf16.

SFT. Each of the three variants trains the Qwen2.5-Omni-7B thinker with all parameters unfrozen (no LoRA) on six H200 GPUs, at a per-GPU batch size of 28 (global batch 168), for two epochs, with

Table 2: Top-1 accuracy on the development set and on the official ADQA-Bench evaluation set (3,000 items, scored by the organizers after the challenge), with the parse rate on dev (share of items whose output the scorer can map to one of the four options, via the answer tag or fallback text heuristics; unmapped items are scored wrong). “subm.” is the submission index; rows marked *ref* are baselines and were not submitted, so they have no official evaluation score; rows marked *post* were not submitted either, and we scored them ourselves with the gold labels released after the challenge. Bold marks our best submitted system.

system	subm.	dev (%)	eval (%)	parse (%)
base Qwen2.5-Omni-7B	ref	53.08	–	86
SFT-A, baseline	ref	55.20	–	97
SFT-B, Q1+Q2	4	55.51	48.33	97
SFT-C, baseline+Q2	3	56.32	48.67	96
GRPO on C, with judge, 300	2	57.31	49.83	99
GRPO on C, no judge, 300	1	58.43	50.60	99
<i>post-challenge runs: no judge, 300 steps, same prompts</i>				
GRPO on C, 2nd seed	post	57.19	50.07	98
GRPO on A	post	56.88	49.73	100
GRPO on A, 2nd seed	post	57.62	50.90	99

maximum sequence length 1024, learning rate 10^{-6} , a cosine schedule, and 5% warmup.

GRPO. Starting from the SFT-C checkpoint, we run on four H200 GPUs with vLLM-served colocated rollouts: 8 rollouts per prompt, sampling temperature 1.5 with top- k 4, one optimization pass per batch, KL coefficient $\beta = 10^{-3}$, learning rate 10^{-6} cosine, maximum completion length 768, and maximum sequence length 4096. We run 600 steps, evaluate the 300- and 600-step checkpoints on the development set, and submit the dev-optimal step-300 checkpoint. The post-challenge runs use the same settings and the same prompts in the same order, and are stopped at step 300.

Inference. All systems decode greedily with repetition penalty 1.05, a 512-token thinker budget, the KV cache enabled, FlashAttention-2, and batch size 16. Top-1 accuracy is scored by extracting the final answer tag and matching it to one of the four options.

Datasets. The official 1,607-item development set, DCASE2026-Task5-DevSet [9], draws a small portion from the MMAU [10], MMAR [11], and MMSU [12] benchmarks, with the majority newly constructed, human-annotated multiple-choice questions; it carries gold labels and is our sole basis for model selection. The official ADQA-Bench evaluation set has 3,000 items whose gold labels were withheld during the challenge and released afterwards; we use them only to score finished systems, never for model selection.

5. RESULTS

Development set. Table 2 reports top-1 accuracy on the full 1,607-item development set and on the official evaluation set, together with each system’s parse rate. Every fine-tuned variant improves over the untuned base model, and accuracy rises at each pipeline stage: SFT lifts the base model by +2.1 to +3.2 points, with or without the repair, and GRPO on top of SFT-C adds up to +2.1 more. The best system is GRPO on variant C *without* the judge term, stopped at 300 steps: **58.43%**, +5.35 points over base (McNemar vs. its SFT-C starting point, $p = 0.057$). The three SFT variants are statistically indistinguishable from one another (paired McNemar, all $p > 0.2$), and only SFT-C’s gain over the base model is significant ($p = 0.015$), so we do not read the SFT ordering as a ranking.

GRPO from the unrepaired model. After the challenge we ran the same GRPO recipe, on the same prompts, from the unrepaired SFT-A, and repeated both initializations with a second seed (Table 2, bottom). Averaged over seeds and both sets, the two initializations differ by 0.2 points (paired permutation test, $p = 0.59$), and which one is ahead

changes with the seed, so the repair does not change accuracy after GRPO either. The GRPO gain itself is robust: +1.6 points over SFT on average ($p < 0.001$), the same from either initialization, and every GRPO run scores above every SFT model on both sets.

The judge reward does not help. At the dev-optimal 300 steps, dropping the Prometheus term gives 58.43% against 57.31% with it (statistically tied, McNemar $p = 0.23$), and 50.60 against 49.83 on the evaluation set. Both gaps are the size of the seed-to-seed spread of the no-judge recipe (Table 2), so the judge adds no measurable accuracy. The judge also dominates step time, so the no-judge run trains $2.4\times$ faster.

Over-training. Both GRPO runs peak at 300 steps and decay by 600 (58.43 to 57.13% without the judge, 57.31 to 56.88% with it). The decay appears under both rewards, so it is a property of prolonged GRPO on this pool rather than of the judge.

Post-challenge ablations. Over the submitted no-judge run, 72.5% of rollout groups produced identical rewards for all 8 rollouts, so they contribute no gradient. After the challenge we re-ran GRPO from the same SFT-C checkpoint in three arms: (i) the plain accuracy+format reward; (ii) the same plus DAPO dynamic sampling [13], which resamples zero-variance groups to restore gradient; and (iii) DAPO plus the judge under a stricter absolute rubric that penalizes assertive but unsupported verdicts. At checkpoints 150/300/450/600 the arms score (i) 58.06/57.44/56.13/56.44, (ii) 57.81/57.25/55.82/56.07, and (iii) 57.87/57.50/57.69/58.12. DAPO does not improve accuracy, so more gradient alone does not help. The judge arm is the only one still improving at step 600 and the most stable across checkpoints (a 0.4-point range, against about 2 points for the judge-free arms). Yet no arm beats the submitted no-judge 58.43% (arm (i) repeats its recipe and reaches 57.44% at step 300, the run-to-run spread also seen in Table 2): the improved judge buys late-training stability, not a higher peak.

What the models write. Accuracy hides how differently the systems reason, so we show two examples (excerpts, in italics).

SFT: the repair removes the transcript (eval_0356)

Question: Which speaker clip corresponds to the same individual as speaker clip 1?

SFT-A (wrong): *“[...] After cross-referencing these with the text, I confirmed that the second speaker’s lines were indeed present in the text.”*

SFT-C (correct): *“I’ve been analyzing the audio clips, focusing on the voice characteristics [...] the third clip’s voice matches perfectly.”*

Applying the Q2 rule of Section 2 to the models’ answers, SFT-A cites a non-existent transcript in 6.9% of evaluation-set answers and SFT-C in 0.6% (7.1% and 0.6% on the development set; McNemar $p < 10^{-30}$). Accuracy is unchanged, even on the items where SFT-A cites a transcript, and looser wordings the rule does not cover (“the passage”, “the script”) fall only from about 13% to 10%. The other robust effect, format compliance, comes from fine-tuning itself, not from the repair: the base model’s output can be mapped to an option on only 86% of items, and only 57% carry a well-formed tag with the exact option text; every fine-tuned system, including the unrepaired SFT-A, reaches a 96 to 100% parse rate, recovering a block of items that would otherwise score zero.

GRPO: one training question at step 296 (all three answers correct)

Question: What best describes the speaker’s gender and emotional delivery?

From C, no judge: *“I hear a female voice speaking with a moderate pitch and a natural, slightly surprised tone. [...]”*

From C, with judge (judge score 5 of 5): *“I’m absolutely locked onto the audio’s core [...] the others simply do not resonate. The analysis is conclusive [...]”*

From A, no judge: *“ANALYSIS OF SPEAKER’S CHARACTER”, a heading and then the answer.*

Without the judge the rationale shrinks during GRPO, in one run to a bare heading (a median of 3 words). With the judge it keeps its length (a median of 70 words at step 300) but grows more confident, not more informative: words such as “absolutely” or “unequivocal” appear in 39% of training samples at the start and in 99.7% at step 300, against 1 to 17% in the four runs without the judge. The judge’s top score goes to 28% of correct answers at the start and to 94% at the end. The model learned what a text-only judge scores highly, a form of reward hacking, and accuracy did not improve.

6. LIMITATIONS AND FUTURE WORK

Teacher reasoning quality. Our regenerated traces come mostly from the compact Gemma-4-E4B-it teacher, with a minority from the larger Qwen3-Omni-30B; both are smaller than the Gemini 3.1 Pro model that wrote the original CoT. The repair removes the harmful signal, but it brings no measurable accuracy gain (variant B lifts SFT only 55.20 to 55.51% over the baseline on dev), so our teachers’ reasoning, while grounded, may not match Gemini’s quality. We did not test a stronger teacher, so whether it would turn the repair into an accuracy gain is open; a stronger teacher pool is the natural next step.

Temporal traces. We could not tell from surface form whether a timestamp was genuinely heard or narrated from a transcript, so we left temporal evidence unflagged; some residual contamination on temporal items may remain.

The judge. The Prometheus reward did not improve accuracy and cost $2.4\times$ the time. Even at its capped weight of 0.2 it changed how the model writes, toward what a text-only judge scores highly rather than toward better answers (Section 5); a stricter rubric steadied training but did not raise the peak, so an audio-aware grader is a promising direction for future work.

7. CONCLUSION

We treated DCASE 2026 Task 5 as a data-quality problem: a deterministic audit showed that most official Gemini 3.1 Pro CoT labels ignore the answer options or reason from a non-existent transcript, the behaviours the benchmark penalizes. Regenerating the flagged traces with a blind, audio-only teacher cascade under rejection sampling yields grounded, option-discriminating supervision, using only the provided data and a sub-10B model for the lightweight track. Our best system reaches 58.4% on dev and 50.6% on the evaluation set with GRPO. In a controlled comparison the repair cuts transcript-citing answers from 6.9% to 0.6% but leaves accuracy unchanged, before and after GRPO, whereas GRPO adds about 1.6 points from either initialization. We therefore recommend auditing and repairing CoT data to keep non-existent sources out of the model’s reasoning, and GRPO for accuracy. A controlled ablation shows the judge adds no measurable accuracy at $2.4\times$ the cost, so our strongest model uses a simple accuracy-and-format reward.

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